Semester 5 · Mechanical / Manufacturing · 4 Credits · 60 periods

Design of Machine Elements

Handbook for mechanisms, fasteners, welded and riveted joints, shafts, keys, cams, flywheels, governors, couplings, bearings and mechanical power transmission drives.

Course meaning

Machine design converts force, motion, material and safety requirements into practical components. This lesson connects mechanism identification, strength design, drawing profile construction and transmission selection.

Malayalam: Machine element design-□ dimension guess □□□□□□□□□. Load, stress, factor of safety, material, failure mode, manufacturing method □□□□□□ □□□□□□ design □□□□□□□.

Official details

Code	5021
Title	Design of Machine Elements
Category	Program Core / Elective
Prerequisites	Mathematics, Engineering Mechanics, Strength of Materials

CO1

Mechanisms, fasteners, welded and riveted joints.

CO2

Shafts, power transmission and keys.

CO3

Cams, flywheels, governors, couplings and bearings.

CO4

Belt, chain, rope and gear drives.

Engineering roadmap

CO1 · 16 Hours

Mechanisms, Fasteners and Welded Joints

Machine design basics

Machine design starts from function, load, material, manufacturing method and safety. Design stress is chosen below material strength using factor of safety. Working stress must not exceed allowable stress.

Mechanisms

A kinematic link transmits motion; a kinematic pair constrains relative motion; a kinematic chain forms a mechanism when one link is fixed. Four bar chain, single slider crank chain and double slider crank chain are important inversions.

Fasteners and welds

Permanent fasteners include welded and riveted joints. Temporary fasteners include threaded joints. Welded joints may be lap/fillet or butt joint. Strength depends on throat area and permissible shear stress.

Throat thickness of fillet weld ≈ 0.707 × leg size.

Expected questions

1. Define link, pair, chain, mechanism and inversion.
2. Classify kinematic pairs and constrained motion.
3. Explain transverse and parallel fillet welds.
4. Identify riveted joint types with sketches.

CO2 · 13 Hours

Shafts and Keys

Shaft design

Shafts transmit power and may be solid or hollow. Design considers torsion, bending, rigidity, material and standard sizes. Hollow shafts can give better strength-to-weight ratio.

Power transmitted $P = 2\pi NT/60$.

For solid circular shaft, $T/J = \tau/R$.

Keys

Keys prevent relative rotation between shaft and hub. Sunk keys, square keys and rectangular keys are common. Key is checked for shear and crushing.

Worked example

If torque $T = 120$ N-m and speed $N = 600$ rpm, power $= 2\pi NT/60 = 7.54$ kW approximately.

Power ≈ 7.54 kW.

Expected questions

1. List shaft types and materials.
2. Compare solid and hollow shafts.
3. Design a shaft on strength basis.
4. Design square and rectangular sunk keys.

CO3 · 14 Hours

Cams, Flywheels and Governors

Cams and followers

A cam gives specified follower motion. Follower motion may be uniform velocity, SHM or uniform acceleration and retardation. Students must draw disc cam profiles for knife-edge and roller followers, with or without offset.

Flywheel and governor

Flywheel stores energy and controls speed fluctuation within one cycle. Governor controls mean speed by regulating input. Watt and Porter governors are centrifugal governors.

Couplings and bearings

Couplings connect shafts. Bearings support rotating shafts and reduce friction. Correct selection depends on load, speed, alignment and maintenance.

Expected questions

1. Draw cam displacement diagrams.

2. Explain turning moment diagram and flywheel function.
3. Compare flywheel and governor.
4. Classify couplings and bearings.

CO4 · 15 Hours

Mechanical Power Transmission Devices

Belt, chain, rope and gear drives

Power can be transmitted by belts, chains, ropes and gears. Belts suit medium distance drives; chains avoid slip; ropes handle large power; gears give positive velocity ratio.

Belt drive calculations

Flat and V-belts are used in open and crossed layouts. Important terms include angle of lap, slip, creep, tight side tension, slack side tension, centrifugal tension and initial tension.

Velocity ratio without slip = driver diameter / driven diameter for belt speed relation.

Gear drives

Gear train value and velocity ratio are used for simple and compound gear trains. Spur gear terminology includes pitch circle, module, addendum, dedendum, pressure angle and number of teeth.

Expected questions

1. Compare belt, chain, rope and gear drives.
2. Solve belt length for open and crossed belts.
3. Explain slip and creep.
4. Solve gear train speed/teeth problems.

Formula bank

Design stress = yield or ultimate strength / FOS. Weld throat = 0.707s. $P = 2\pi NT/60$. Shaft torsion relation $T/J = \tau/R$. Module m = pitch circle diameter / teeth.

Model paper

1. Explain kinematic chain and inversions.
2. Solve shaft torque and power problem.
3. Draw cam profile steps for roller follower.
4. Compare belt, chain, rope and gear drives.

Practice MCQ

1. Flywheel controls cyclic speed fluctuation. 2. Governor controls mean speed. 3. Chain drive is positive drive. 4. Throat thickness of fillet weld is 0.707 times leg size.

Answer guide

For design answers show assumption, load, material, factor of safety, formula, substitution, final dimension and practical check. For diagrams use standard labels. For comparison answers use table format.

References: S.S. Rattan, R.K. Bansal, Sadhu Singh, V.B. Bhandari, R.S. Khurmi and J.K. Gupta, and NPTEL resources.